Jordan Watkins, PhD

Curriculum Vitae — mathematician, software engineer, educator

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Mathematician and software engineer with a PhD in differential geometry and a decade of production experience. Published original research on rank rigidity; co-created and shipped a puzzle game rated 98% positive on Steam; architected a Python codebase that has run two community services continuously for ten years; and built a Bayesian rating and matchmaking system validated against held-out season data. Four years of university teaching, plus tutoring across mathematics, physics, and computer science.

EDUCATION

Ph.D. in Mathematics · University of Michigan — 2013

Specialization: differential geometry (dynamics on Riemannian manifolds). Thesis: "The Rank Rigidity Theorem for Manifolds With No Focal Points."

B.A. in Mathematics and Physics · Knox College — 2007

Valedictorian

PUBLICATIONS

"The Higher Rank Rigidity Theorem for Manifolds With No Focal Points" · *Geometriae Dedicata*, vol. 164 — Aug 2012 · pp. 319–349

Springer · arXiv:1111.5428

RESEARCH

Doctoral thesis · University of Michigan — 2013

"The Rank Rigidity Theorem for Manifolds With No Focal Points" — an extended treatment of the published higher-rank rigidity result.

SOFTWARE & QUANTITATIVE WORK

Co-Developer — *Bean and Nothingness* · RBOR Games — 2020–2021 · shipped Dec 2021

store.steampowered.com/app/1706090/Bean_and_Nothingness/

Tile-based logic-puzzle game · 98% positive from 198 Steam reviews · 180+ puzzles · Steam Workshop support

- Co-lead programmer (C++, SDL, OpenGL); deeply familiar with nearly the full codebase. Grew out of prototype work begun in 2012
- Designed and implemented a replay / undo / state-saving system requiring precise modeling of game state across arbitrary action sequences
- Built a GUI level editor used internally and shipped publicly via Steam Workshop
- Implemented a custom sound mixer and contributed to the core game loop and entity logic architecture
- Lead puzzle designer: created 100+ puzzles, edited all ~200, and owned difficulty progression across the whole game
- Coordinated a distributed remote team from prototype through release — scoping, sequencing content, and driving the project to ship

Competitive Rating & Matchmaking Systems · CoNDOR · open source

multiple seasons, through 2022

github.com/incnone/condor_ratings

A Bayesian rating model, series-outcome forecasting, and automated matchmaking, rebuilt and refined across several league seasons (Python, C++, MySQL, PuLP)

- Built and iterated on a Bayesian paired-comparison rating model: per-player Gaussian priors combined with binomial match evidence, fit by MAP estimation
- Derived the log-likelihood gradient analytically and wrote the optimizer by hand — batch gradient ascent with L2 convergence tests on both parameter change and gradient norm, iteration and wall-clock limits, and log-likelihood tracing
- Selected the prior variance by maximizing log-likelihood on held-out matches from a previous season — model selection against real data rather than hand-tuning
- Judged a normal model's tails too thin for real performance variation and designed a heavier-tailed link function — a mixture of an error function and a rational sigmoid whose tail weight is a hyperparameter fit to past match data ($\alpha = 0.42$) — calibrated to agree with the Bradley–Terry model at a chosen reference rating
- Patched Rémi Coulom's C++ Bayesian Elo engine (bayeselo) to expose the covariance matrix it computes internally, then modeled two players' ratings as jointly normal so series predictions account for correlated rating uncertainty
- Converted ratings into calibrated win probabilities and full best-of-3 / best-of-5 outcome distributions, exported as JSON for downstream use
- Automated weekly matchmaking as an integer linear program (PuLP/CBC) maximizing an information-theoretic objective — the entropy of the predicted result, penalized by rating gap — under per-player degree constraints
- Added a local-search pass performing minimum-cost pairwise exchanges to enforce a minimum cycle length in the matchup graph, with unit tests

Architect & Maintainer — Necrobot · Open source

2015–Present

github.com/incnone/necrobot

Discord bots managing competitive speedrunning races and league seasons in *Crypt of the NecroDancer*, in continuous community use for 10+ years

- Architected one codebase that runs two distinct production bots (Necrobot and Condorbot) — ~150 modules across 16 packages with a documented, acyclic package dependency order
- Built an internal command framework separating input parsing and user-facing error handling from data manipulation, over an object layer mapped to MySQL tables
- Async Python (asyncio, aiohttp, discord.py) with MySQL persistence, Google Sheets integration for standings, and timezone-aware match scheduling
- Race ratings built on TrueSkill (Bayesian), adapted to the race format of the game
- Lead architect and maintainer across 10+ years, coordinating a small contributor team; MIT-licensed with admin, code, and database documentation

TEACHING EXPERIENCE

Graduate Student Instructor, Mathematics · University of Michigan

2007–2011

- Independently taught multiple sections of Calculus I and Calculus II — lecturing, assessment, and grading
- Teaching assistant for Differential Equations
- Led required weekly tutoring sessions in single- and multivariable calculus, differential equations, and basic probability
- Mentored individual paying students, including in linear algebra

Mathematics & Physics Tutor · Independent

Intermittently since 2005

- Tutored across a wide range: topology, abstract algebra, differential equations, statistics, linear algebra, and electricity & magnetism
- Supported computer-science students in C++
- Graduate-school teaching included weekly tutoring in calculus, differential equations, and probability, plus independent paying students in linear algebra

SUBJECTS TAUGHT & TUTORED

Calculus I–III · Differential equations · Linear algebra · Abstract algebra · Topology · Probability & statistics · Electricity & magnetism · C++ · Python

INDEPENDENT WORK

Independent Study & Freelance Tutoring

2013–2020

- Self-directed study across mathematics, physics, machine learning, and data science
- Occasional freelance remote tutoring in mathematics and physics
- Ongoing open-source development and community leadership (Necrobot, from 2015)
- Early prototype work on the game that became *Bean and Nothingness*

CONTINUED EDUCATION

Mathematical Physics Seminar · University of Virginia

2024–Present

Regular participant. Ongoing engagement with current research in geometry and mathematical physics.

Erdős Institute — Data Science Boot Camp · coursework

Applied data science in Python (pandas, scikit-learn).

Self-directed study in machine learning (PyTorch), AI ethics, and post-doctoral mathematics and physics; ~4,000 reputation and 86 answers on Mathematics Stack Exchange.

SERVICE & COMMUNITY

- Architect, maintainer, and community moderator for the open-source Necrobot project and its competitive community (10+ years)
- Rating and matchmaking systems administered across multiple league seasons
- Policy and moderation on large public Discord servers

SKILLS

Languages	C++, Python, SQL
Systems & tools	SDL, OpenGL, asyncio/aiohttp, discord.py, MySQL, PuLP, Google Sheets API, Git, Visual Studio
Mathematics	Differential geometry, Bayesian modeling, optimization, probability, information theory
Practice	Engine and service architecture, tooling, project leadership, teaching, long-term maintenance

Career note: independent study and freelance tutoring 2013–2020, during part of which I carried family caregiving responsibilities.